Automate Cisco ACI

with Network as Code (NaC)

Human-Readable Intent

Declarative Configuration

```yaml
apic:
  node_policies:
    nodes:
      - id: 101
        pod: 1
        role: leaf
        serial_number: ABC1234567
        name: LEAF101

      - id: 1001
        pod: 1
        role: spine
        set_role: true
        serial_number: ABC1234569
        name: SPINE1001
```

Write intent in simple, human-readable YAML.

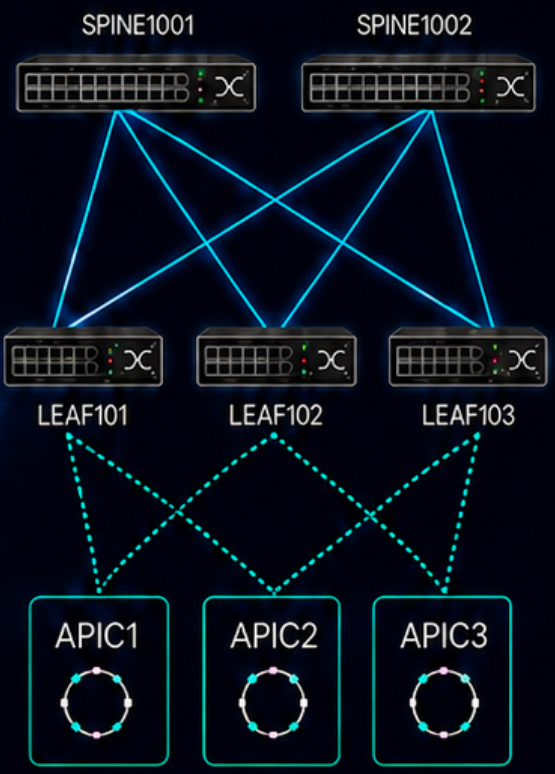
Network as Code

Terraform

Terraform transforms intent into ACI policy configuration.

Policy-Driven ACI Fabric

Automated & Consistent



Author: Titus Majeza

Introduction to Network as Code (NaC)

Cisco Network as Code (NaC) is an automation approach that applies Infrastructure as Code (IaC) principles to network infrastructure. It uses declarative, human-readable data models to define the intended state of routers, switches, firewalls, and software-defined networks.

By representing network intent as code, configurations can be version-controlled, validated, tested, and deployed through automated workflows and CI/CD pipelines. This enables network changes to be delivered in a more consistent, repeatable, and auditable manner.

> Infrastructure as code at a high level is the practice of provisioning and managing infrastructure through human-readable declarative data models rather than through manual processes.

Network as Code for ACI

For many Cisco ACI deployments today, infrastructure is commonly provisioned through the APIC graphical user interface (GUI), REST API calls using tools such as Postman, automation frameworks like Ansible, or custom Python scripts. Each of these approaches has its place and continues to be valuable depending on the operational requirements.

The APIC GUI, in particular, provides an intuitive way to learn Cisco ACI, visualize policy relationships, and perform day-to-day operational tasks. However, as deployments grow from a handful of objects to hundreds or even thousands of tenants, VRFs, Bridge Domains, EPGs, contracts, and external connections, the operational challenge shifts. Engineers spend increasing amounts of time navigating between screens, creating dependent objects in the correct order, resolving validation faults after deployment, and manually ensuring that configuration remains consistent across environments.

Provisioning through the GUI can require extensive navigation across different areas of the APIC, where related objects must be created and connected in the correct sequence. Errors may only become visible after configuration has already been submitted, often through faults that engineers must investigate and resolve reactively. Meanwhile, isolated Postman collections, Ansible playbooks and Python scripts can gradually become separate automation islands, each managing only part of the infrastructure and potentially maintaining its own understanding of the network state.

The result is a familiar operational problem: manual processes (prone to human error), multiple tools, multiple configuration touchpoints, no single authoritative representation of the intended network, no record of network changes (lack of version control) and lack of extensive testing and validation.

This is where Network as Code changes the operational model. Rather than treating infrastructure as a collection of individual configuration tasks/actions, Network as Code allows the desired state of the network to be described declaratively, through a simplified, human-readable data model, validated before deployment, version-controlled, peer-reviewed, and deployed through repeatable automation pipelines.

The objective is not merely to configure ACI faster. It is to build a disciplined engineering lifecycle in which every proposed change can be reviewed, validated before deployment, applied consistently and tested afterwards to confirm that the declared intent has been pushed to the infrastructure and that the infrastructure has reached its intended operational state.

Terraform/OpenTofu can be used to serve as the automation engines which render the intended state defined by the user, into the actual API calls that will be sent to the network infrastructure (in this case; the APIC).

Author: Titus Majeza

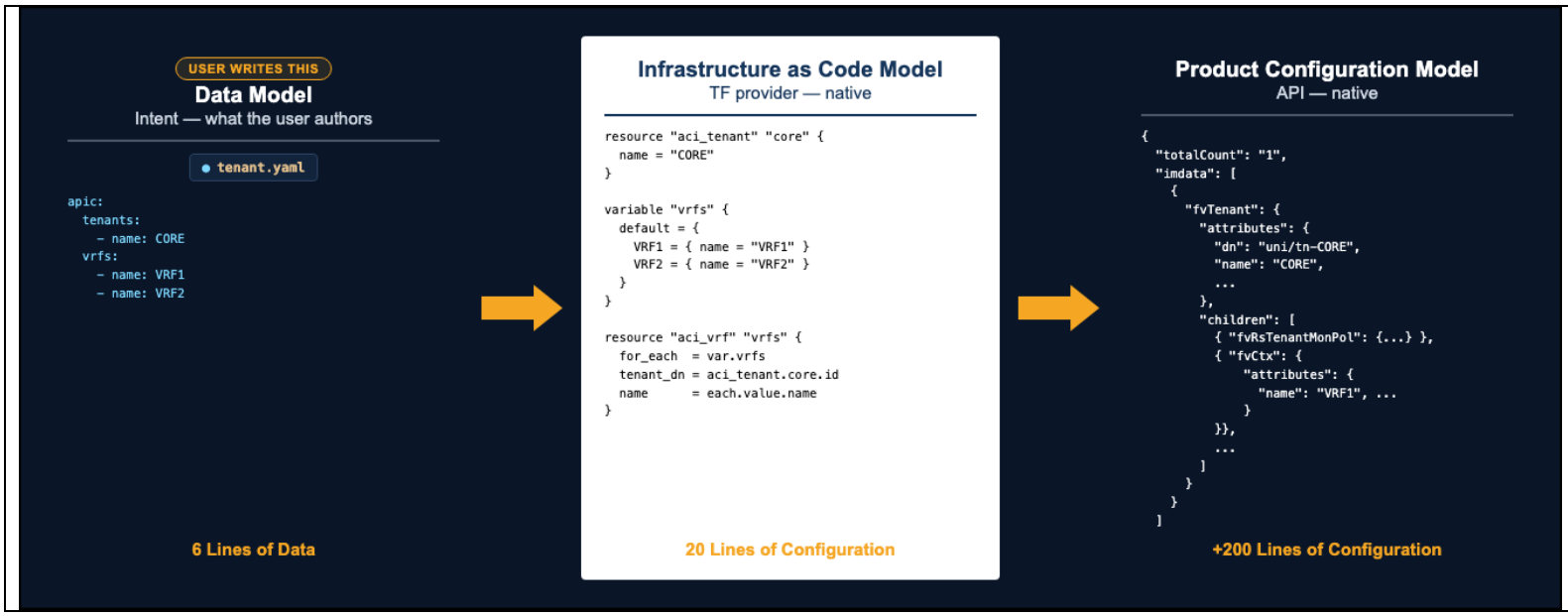
Network as Code – Abstraction

Cisco ACI already provides a rich and highly programmable object model. Every tenant, VRF, Bridge Domain, EPG and contract is represented through managed objects that can be accessed through the APIC REST API.

This flexibility is powerful, but it also exposes engineers to a large amount of implementation detail. Direct API automation requires an understanding of object classes, distinguished names, parent-child relationships, attributes and dependencies.

The native Terraform provider simplifies this interaction by representing ACI managed objects as Terraform resources. However, engineers may still need to define multiple resources, references, loops and dependencies to express what is ultimately a relatively simple network requirement.

Cisco Network as Code introduces another layer of abstraction. Instead of defining every underlying ACI object individually, engineers describe the desired network configuration using a simplified YAML data model. The Network as Code Terraform module then translates that intent into the required ACI resources and API operations. This reduces time to value by abstracting away scripting syntax and low-level device configuration behind a curated, opinionated data model, so users can focus entirely on describing what they want configured.



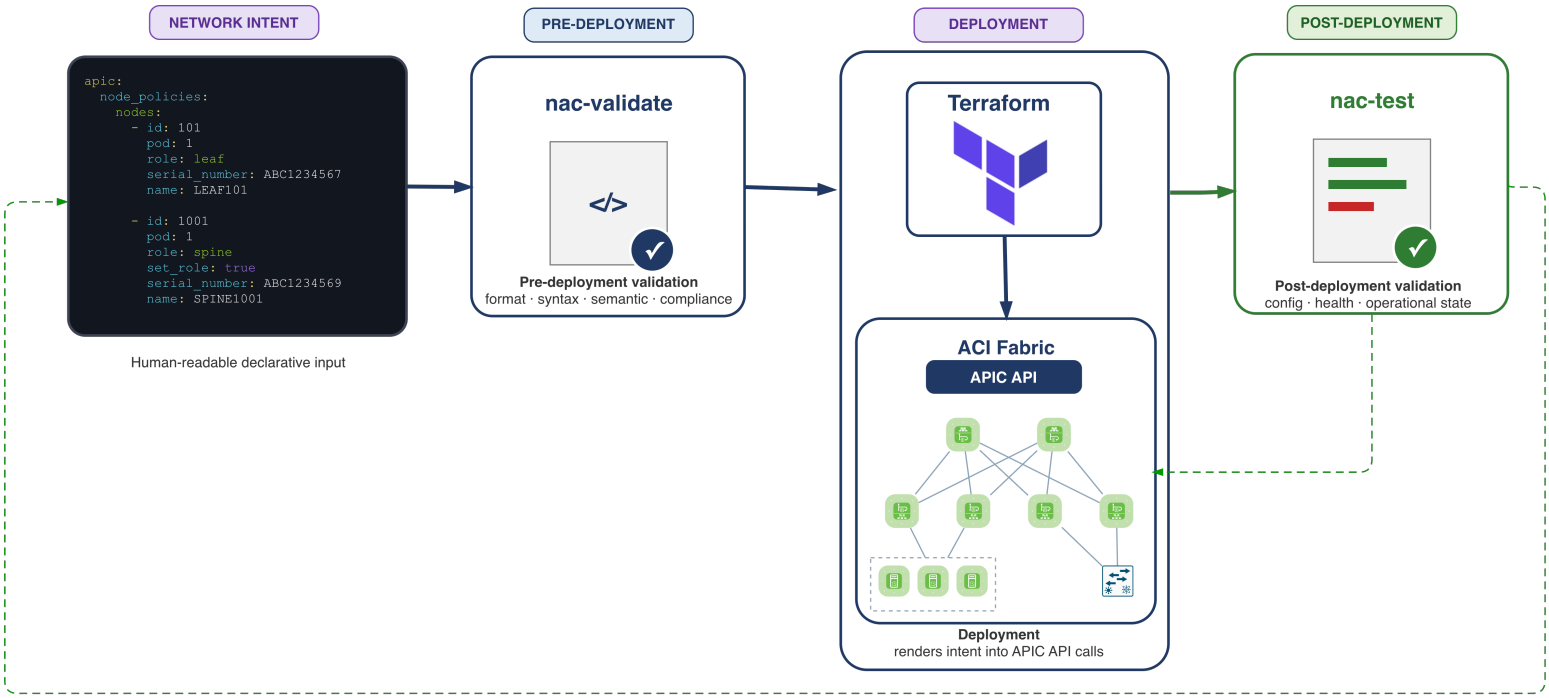
Lab Setup

This lab consists of a two-spine, four-leaf Cisco ACI fabric in its initial, unconfigured state. Network as Code will be used to build and configure the fabric from scratch, transforming declarative network intent into an operational ACI environment.

Network as Code Implementation

The workflow below demonstrates the stages involved in delivering ACI deployment and validation using Network as Code. This workflow illustrates an end-to-end **Network as Code (NaC)** approach for deploying and validating Cisco ACI configuration. Network intent is first defined using a **human-readable declarative data model**, which describes the desired state of the fabric. Before any change is applied, **nac-validate** performs pre-deployment checks to validate the syntax, semantics, and compliance of the intended configuration. Once validation succeeds, Terraform translates the declared intent into the required ACI policy configuration and applies it to the fabric through the APIC API. After deployment, **nac-test** performs post-

deployment validation to confirm that the configuration was successfully implemented and that the fabric is in the expected operational state. Together, these stages create a repeatable validation and deployment workflow in which network intent can be defined, verified, deployed, and tested consistently. Each stage is demonstrated in detail throughout the remainder of this document.



Getting Started

Pre-requisites

- Terraform installed on your development machine

```
nac-aci-terraform % terraform --version
Terraform v1.15.8
```

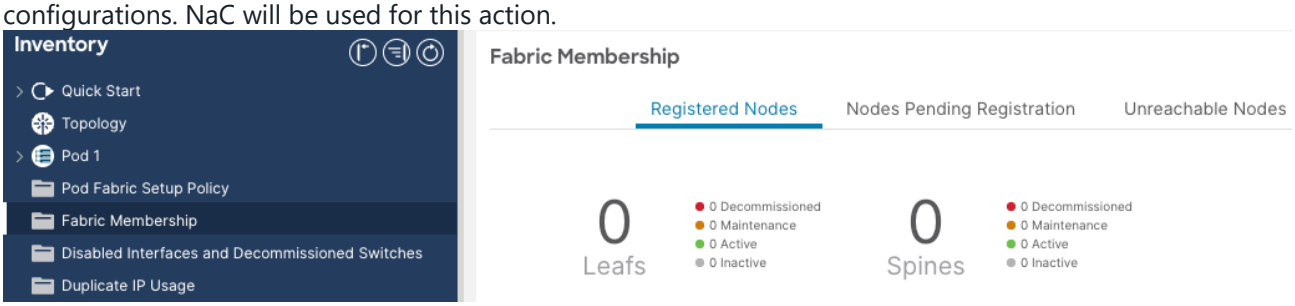
- Python installed on your development machine
    - Python 3.11, 3.12 or 3.13 (required by nac-validate and nac-test)

- nac-validate and nac-test installed on your development machine

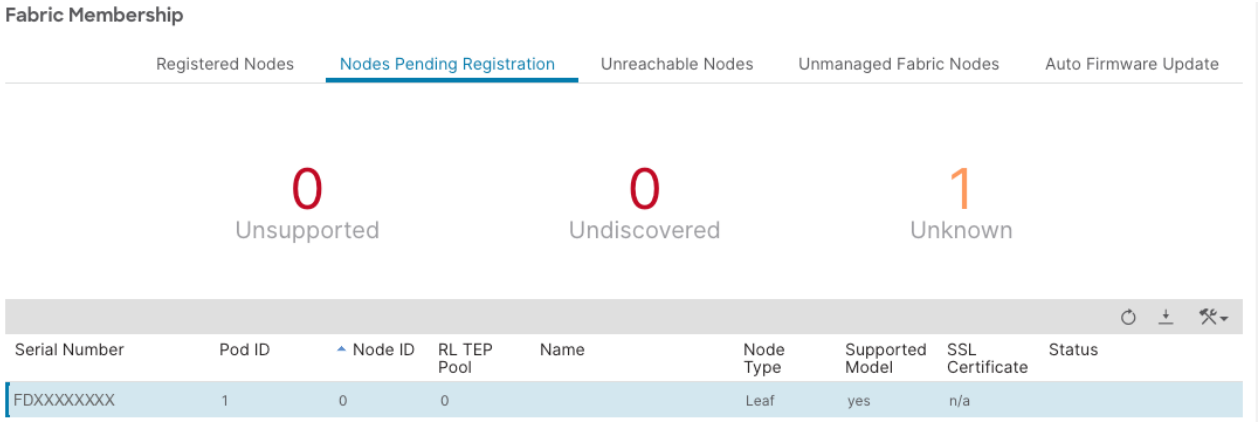
```
pip install nac-validate nac-test
nac-validate --version
nac-test --version
```

- ACI infrastructure (real hardware or simulator)

Currently the ACI infrastructure does not have any nodes registered and does not have any pre-existing configurations. NaC will be used for this action.



Author: Titus Majeza



- Create a project folder in this structure.

```
nac-aci-terraform/
├── main.tf
└── data/
    ├── apic.nac.yaml
    ├── defaults.nac.yaml
    └── node_policies.nac.yaml
```

File	Description
`apic.nac.yaml`	Network as Code (ACI) configuration like enabling/disabling auto-generation of switch/interface policies, etc.
`defaults.nac.yaml`	Network as Code (ACI) defaults used to overwrite the defaults built into the Terraform Modules
`node_policies.nac.yaml`	ACI node registration, node management IP, update group, and vPC protection group configuration

The intent for node registration, out of band assignment, vPC groups configs etc. is expressed in the node_policies.nac.yaml.

```
---
apic:
  node_policies:
    vpc_groups:
      mode: explicit
      groups:
        - id: 1
          switch_1: 1001
          switch_2: 1002
        - id: 2
          switch_1: 1003
          switch_2: 1004

    nodes:
      - id: 1001
        pod: 1
        role: leaf
        serial_number: FDXXXXXXXX
        name: leaf1
        oob_address: 10.x.y.232/27
        oob_gateway: 10.x.y.225
        fabric_policy_group: ALL_LEAFS
        access_policy_group: ALL_LEAFS

      - id: 1002
        pod: 1
        role: leaf
        serial_number: FDXXXXXXXX
        name: leaf2
        oob_address: 10.x.y.233/27
        oob_gateway: 10.x.y.225
        fabric_policy_group: ALL_LEAFS
```

```
    access_policy_group: ALL_LEAFS

  - id: 1003
    pod: 1
    role: leaf
    serial_number: FDXXXXXXXX
    name: leaf3
    oob_address: 10.x.y.234/27
    oob_gateway: 10.x.y.225
    fabric_policy_group: ALL_LEAFS
    access_policy_group: ALL_LEAFS

  - id: 1004
    pod: 1
    role: leaf
    serial_number: FDXXXXXXXX
    name: leaf4
    oob_address: 10.x.y.235/27
    oob_gateway: 10.x.y.225
    fabric_policy_group: ALL_LEAFS
    access_policy_group: ALL_LEAFS

  - id: 201
    pod: 1
    role: spine
    serial_number: FDXXXXXXXX
    name: spine1
    oob_address: 10.x.y.236/27
    oob_gateway: 10.x.y.225
    fabric_policy_group: ALL_SPINES

  - id: 202
    pod: 1
    role: spine
    serial_number: FDXXXXXXXX
    name: spine2
    oob_address: 10.x.y.237/27
    oob_gateway: 10.x.y.225
    fabric_policy_group: ALL_SPINES
```

The YAMLs definition is not random. It is guided by the official data model which can be accessed from here: https://netascode.cisco.com/docs/data_models/ .The figure below shows an example of an ACI node registration data model. A user defining a YAML file can see what objects are supported, mandatory/optional, default values etc.

Classes

nodes *(apic.node_policies)*

Name	Type	Constraint	Mandatory	Default Value
role	Choice	`leaf` , `spine` , `apic`	Yes	
set_role	Boolean	`true` , `false`	No	`false`
serial_number	String	Regex: `^[a-zA-Z0-9_.:-]{1,16}$`	No	
name	String	Regex: `^[a-zA-Z0-9_.:-]{1,64}$`	No	
id	Integer	min: `1` , max: `4000`	Yes	
pod	Integer	min: `0` , max: `255`	No	`1`
type	Choice	`tier-2-leaf` , `remote-leaf-wan` , `virtual` , `border-gateway`	No	
remote_pool_id	Integer	min: `1` , max: `255`	No	
expected_state	Class	`[expected_state]`	No	

The ***nac-aci*** module is the key centerpiece that ties the data in the YAML file to the terraform provider for ACI that will be used to configure ACI. The module defines the terraform code, required logic, loops, resource dependencies which translates the data defined in the YAML file into the respective APIC calls for ACI automation.

This module is publicly available in the terraform registry:
https://registry.terraform.io/modules/netascode/nac-aci/aci/latest

The module is defined in the main.tf file

```
module "aci" {
  source  = "netascode/nac-aci/aci"
  version = "2.0.0"

  yaml_directories = ["data"]

  manage_access_policies    = false
  manage_fabric_policies    = false
  manage_pod_policies       = false
  manage_node_policies      = true
  manage_interface_policies = false
  manage_tenants            = false

  write_default_values_file = "defaults.yaml"
}
```

As seen from the main.tf file, the directory where the YAML files lie is defined, along with the specific objects that the user wants to manage using terraform. For now node_policies have been marked to true which means anything else that lies outside of the node_policies data model will not be considered for provisioning.

Now that the data is ready, provide infrastructure credentials (in this case APIC) as environment variables. These variables are required by terraform to communicate with the ACI infrastructure.

```
export ACI_USERNAME=admin
export ACI_PASSWORD='APIC_PASSWORD'
export ACI_URL=https://APIC-IP
```

Instead of exporting credentials in your shell, they can be retrieved from a secrets manager such as HashiCorp Vault and injected as environment variables at runtime, so nothing is left in your shell history. Values in the YAML data model can then reference them with the `!env` tag (for example, `password: !env APIC_PASSWORD`). Note that an unset variable resolves to an empty string rather than raising an error.

So far, the development environment just has the files (YAML and main.tf) but without the modules installed the environment has no capabilities to push configurations to ACI. Hence, the next step is to initialize Terraform to download all the required provider modules that are required for execution.

Initialize terraform

terraform init

This will pull the required terraform providers along with the nac-aci module code from the registry.

```
nac-aci-terraform % terraform init
Initializing the backend...

Initializing modules...
Downloading registry.terraform.io/netascode/nac-aci/aci 2.0.0 for aci...
- aci in .terraform/modules/aci
…

Initializing provider plugins...
- terraform.io/builtin/terraform is built in to Terraform
```

```
- Finding hashicorp/local versions matching ">= 2.3.0"...
- Finding ciscodevnet/aci versions matching ">= 2.0.0, >= 2.5.0, >= 2.13.0, >= 2.15.0, >= 2.17.0, >=
2.19.0"...
- Finding netascode/utils versions matching "2.0.0"...
…
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.
```

After terraform has been initiliazed, a .terraform folder will appear in the working directory. This folder contains the aci module and all its dependencies required to successfully interpret the YAML data and translate that into actual configuration elements.

```
.terraform/
├── modules/
│   ├── aci/
│   └── modules.json
└── providers/
    └── registry.terraform.io/
        ├── ciscodevnet/
        ├── hashicorp/
        └── netascode/
```

Now it's time to generate a dry run of what will be provisioned to the infrastructure. This is achieved by running the terraform plan command. This command allows terraform to query the actual state of the infrastructure, compares it to what has been defined as the desired state and from there the plan of what objects need to be added/remove/modified such that the infrastructure matches the desired state.

terraform plan

```
nac-aci-terraform % terraform plan

Terraform used the selected providers to generate the following execution plan. Resource actions
are indicated with the following symbols:
  + create

  # module.aci.module.aci_node_registration["1002"].aci_rest_managed.fabricNodeIdentP will be
created
  + resource "aci_rest_managed" "fabricNodeIdentP" {
      + annotation  = "orchestrator:terraform"
      + class_name  = "fabricNodeIdentP"
      + content     = {
          + "extPoolId" = null
          + "name"      = "leaf2"
          + "nodeId"    = "1002"
          + "nodeType"  = "unspecified"
          + "podId"     = "1"
          + "role"      = null
          + "serial"    = "FDXXXXXXXX"
        }
      + dn          = "uni/controller/nodeidentpol/nodep-FDXXXXXXXX"
      + escape_html = true
      + id          = (known after apply)
    }
….
Output truncated …

Plan: 25 to add, 0 to change, 0 to destroy.
─────────────────────────────────────────────────────────────────────────────────────────────────

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take
exactly these actions if you run "terraform
apply" now.
```

From the output we can see the objects that will be created.

After reviewing the planned changes, the "terrafom apply" command will carry out the execution plan to the actual infrastructure.

terraform apply

```
nac-aci-terraform % terraform apply
…output truncated

Plan: 25 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

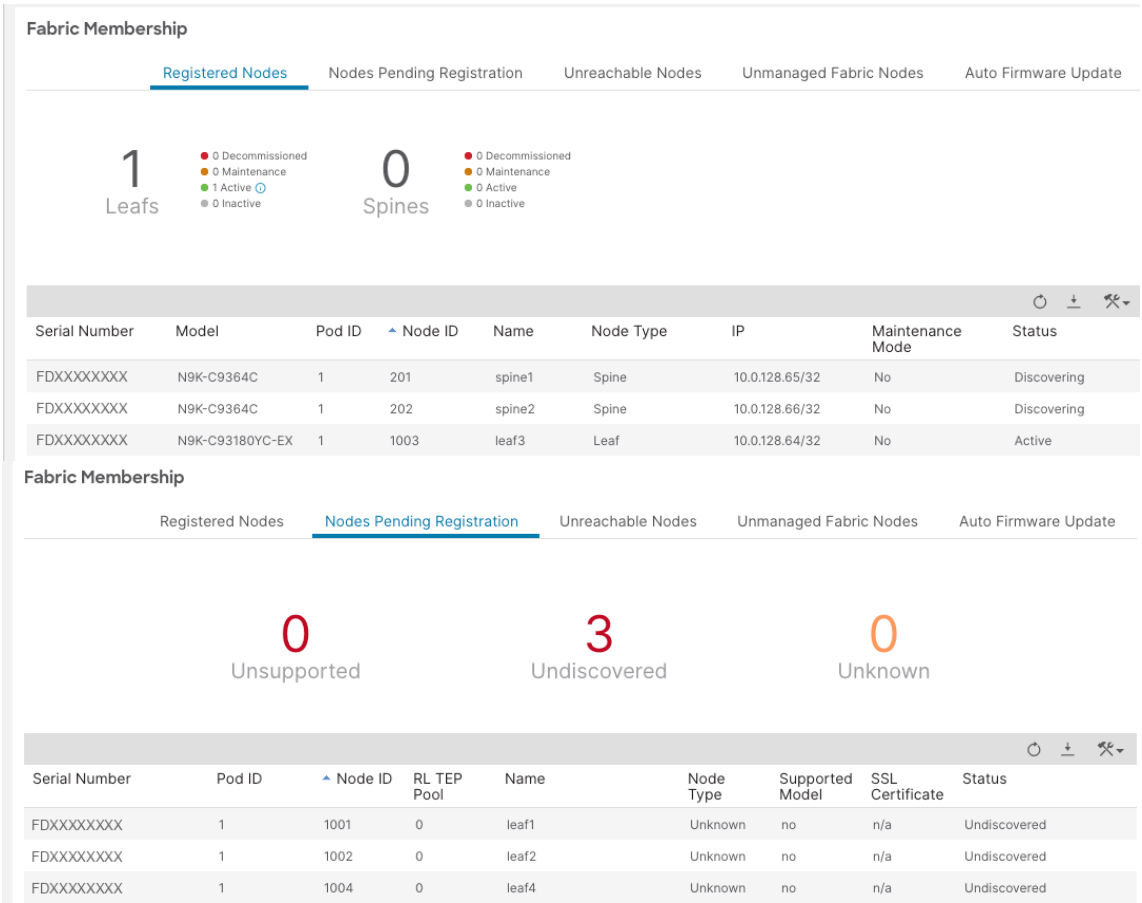
  Enter a value: yes

…output truncated
module.aci.module.aci_oob_node_address["1002"].aci_rest.mgmtOoB: Creation complete after 2s [id={}]
module.aci.module.aci_oob_node_address["202"].aci_rest.mgmtOoB: Creation complete after 2s [id={}]
module.aci.module.aci_oob_node_address["1004"].aci_rest_managed.mgmtRsOoBStNode: Creating...
module.aci.module.aci_oob_node_address["1001"].aci_rest_managed.mgmtRsOoBStNode: Creating...
module.aci.module.aci_oob_node_address["201"].aci_rest_managed.mgmtRsOoBStNode: Creating...
module.aci.module.aci_oob_node_address["1003"].aci_rest_managed.mgmtRsOoBStNode: Creating...
module.aci.module.aci_oob_node_address["1002"].aci_rest_managed.mgmtRsOoBStNode: Creating...
module.aci.module.aci_oob_node_address["202"].aci_rest_managed.mgmtRsOoBStNode: Creating...
module.aci.module.aci_oob_node_address["1004"].aci_rest_managed.mgmtRsOoBStNode: Creation complete after 0s [id=uni/tn-mgmt/mgmtp-default/oob-OOB/rsooBStNode-
[topology/pod-1/node-1004]]
module.aci.module.aci_oob_node_address["1002"].aci_rest_managed.mgmtRsOoBStNode: Creation complete after 0s [id=uni/tn-mgmt/mgmtp-default/oob-OOB/rsooBStNode-
[topology/pod-1/node-1002]]
module.aci.module.aci_oob_node_address["202"].aci_rest_managed.mgmtRsOoBStNode: Creation complete after 0s [id=uni/tn-mgmt/mgmtp-default/oob-OOB/rsooBStNode-
[topology/pod-1/node-202]]
module.aci.module.aci_oob_node_address["1003"].aci_rest_managed.mgmtRsOoBStNode: Creation complete after 0s [id=uni/tn-mgmt/mgmtp-default/oob-OOB/rsooBStNode-
[topology/pod-1/node-1003]]
module.aci.module.aci_oob_node_address["201"].aci_rest_managed.mgmtRsOoBStNode: Creation complete after 0s [id=uni/tn-mgmt/mgmtp-default/oob-OOB/rsooBStNode-
[topology/pod-1/node-201]]
module.aci.module.aci_oob_node_address["1001"].aci_rest_managed.mgmtRsOoBStNode: Creation complete after 0s [id=uni/tn-mgmt/mgmtp-default/oob-OOB/rsooBStNode-
[topology/pod-1/node-1001]]

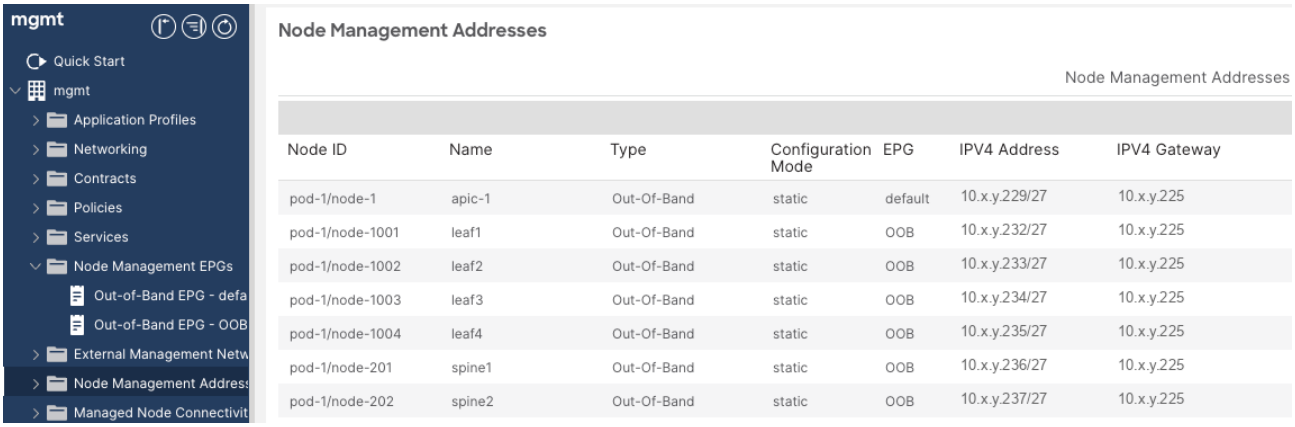
Apply complete! Resources: 25 added, 0 changed, 0 destroyed.
```

After the configuration has been pushed, the node discovery and registration proceess occurs on the APIC.

An operation that would have taken multiple clicks, across different panes, requiring a user to manually input the data for each node, now happens from the execution of a single command.



The nodes also had the out of band management IPs assigned to them.



After the initial configuration has been pushed, additional configuration is done easily by declaring intent in a YAML file and ensuring that the specific ACI modules (node policies, tenants etc.) are marked as true in the main.tf file so that they can be managed by terraform.

Additional YAMLs files have been added to the directory to configure access policies.

```
nac-aci-terraform/
 ├──── main.tf
 └──── data/
      ├──── apic.nac.yaml
      ├──── defaults.nac.yaml
      ├──── node_policies.nac.yaml
      ├──── node_1001.nac.yaml
      ├──── node_1002.nac.yaml
      └──── access_policies.nac.yaml
```

With Network as code, configuration can be split into multiple YAMLs which allows full flexibility for managing configurations. Network as code will perform the deep merging of all the YAML files where configuration is defined. The manage_* flags define which parts of the fabric a Terraform project is responsible for, which helps limit the blast radius of a change: a project that only manages access policies will never plan or modify tenant configuration. Note that these flags set ownership, not a temporary filter. Once a section has been applied, setting its flag back to false removes it from Terraform's scope, and Terraform will plan to destroy the objects it previously created.

Below are the YAML contents required to configure access policies.

main.tf

```
module "aci" {
  source  = "netascode/nac-aci/aci"
  version = "2.0.0"

  yaml_directories = ["data"]
  manage_access_policies    = true
  manage_fabric_policies    = true
  manage_pod_policies       = true
  manage_node_policies      = true
  manage_interface_policies = true
  manage_tenants            = false

  write_default_values_file = "defaults.yaml"
}
```

apic.nac.yaml

```
---
apic:
```

access_policies.nac.yaml

```
apic:
  access_policies:

    vlan_pools:
      - name: LAB_VLANS
        allocation: static
        ranges:
          - from: 3965
            to: 3999
    physical_domains:
      - name: PDOM_LAB
        vlan_pool: LAB_VLANS

    aaeps:
      - name: AEP_LAB_01
        physical_domains:
          - PDOM_LAB
        infra_vlan: false
```

```
  auto_generate_switch_pod_profiles: true
```

node_1001.nac.yaml

```
apic:
  interface_policies:
    nodes:
      - id: 1001
        interfaces:
          - port: 3
            description: "Server Port"
            policy_group: SERVER_PORTS
```

node_1002.nac.yaml

```
apic:
  interface_policies:
    nodes:
      - id: 1002
        interfaces:
          - port: 4
            description: "Server Port"
            policy_group: SERVER_PORTS
```

```
  interface_policies:
    lldp_policies:
      - name: LLDP-ENABLED
        admin_rx_state: true
        admin_tx_state: true
    cdp_policies:
      - name: CDP-ENABLED
        admin_state: true
    link_level_policies:
      - name: 10G-AUTO_ON
        speed: 10G
        auto: true

  leaf_interface_policy_groups:
    - name: SERVER_PORTS
      description: "Server Ports"
      type: access
      link_level_policy: 10G-AUTO_ON
      cdp_policy: CDP-ENABLED
      lldp_policy: LLDP-ENABLED
      aaep: AEP_LAB_01
  leaf_switch_policy_groups:
    - name: ALL_LEAFS
      cdp_policy: CDP-ENABLED
      lldp_policy: LLDP-ENABLED

  spine_switch_policy_groups:
    - name: ALL_SPINES
      lldp_policy: LLDP-ENABLED
      cdp_policy: CDP-ENABLED
```

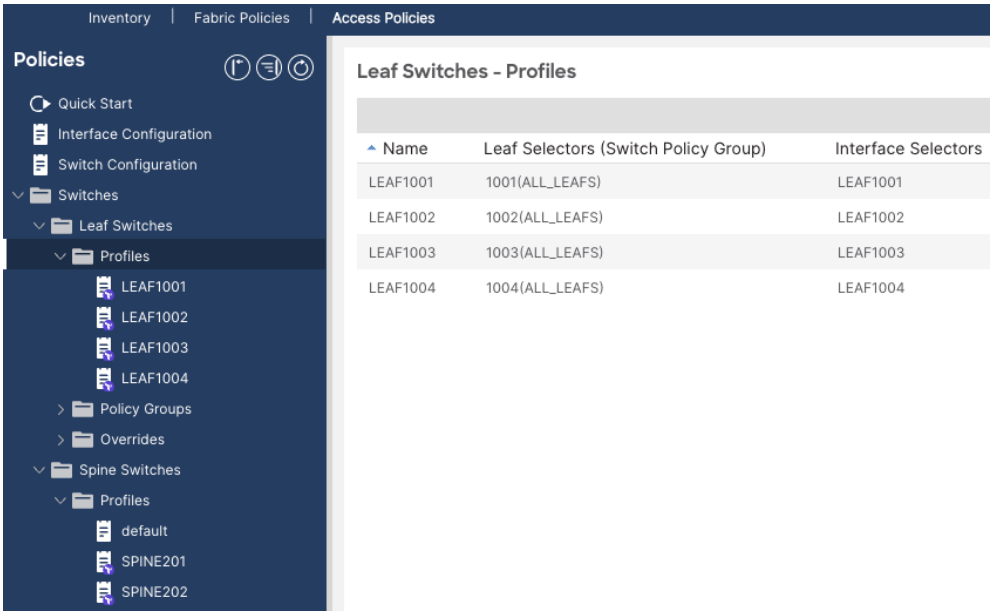
An interesting point to highlight is the `apic.nac.yaml`, which contains a Network as Code based setting which allows for the auto-generation of Leaf/Spine Profiles (Fabric & Access). These are generated automatically using the module's default naming convention (for example, LEAF<node-id> and SPINE<node-id>), so the profiles were never explicitly defined in the YAML files.

The naming patterns come from the module's default values, which Terraform writes to *defaults.yaml* (see the note on defaults.yaml in the Testing and Validation section). The placeholder \g<id> is replaced with the node ID, so node 1001 becomes LEAF1001 and node 201 becomes SPINE201:

defaults.yaml

```
defaults:
  apic:
    access_policies:
      leaf_switch_profile_name: "LEAF\\g<id>"
      leaf_interface_profile_name: "LEAF\\g<id>"
      leaf_switch_selector_name: "LEAF\\g<id>"
      spine_switch_profile_name: "SPINE\\g<id>"
      spine_interface_profile_name: "SPINE\\g<id>"
      spine_switch_selector_name: "SPINE\\g<id>"
```

The profiles below were created because `auto_generate_switch_pod_profiles: true` is set in `apic.nac.yaml`, and their names follow the patterns defined in `defaults.yaml`.



Working with default values

Network as Code ships with sensible built-in defaults based on Cisco's best practices. In some cases, those default values might not be the best choice for a particular deployment and can be overwritten if needed. Appending suffixes to object names is a common practice that introduces room for human errors. Using default values, such suffixes can be defined once and then consistently appended to all objects of a specific type including its references. The same approach works for any attribute that should be identical across many objects. For example, in an L2-only fabric,`unicast_routing` can be disabled once in the defaults instead of being set explicitly under every bridge domain.

To validate how default value overrides behave in practice, a LAB_TENANT tenant is defined in data/lab_tenant.nac.yaml, containing a VRF (LAB_VRF) and three bridge domains (vlan_1001, vlan_1002, vlan_1003) associated with it. Separately, *data/defaults.nac.yaml* overrides two of the module's built-in bridge domain defaults: *name_suffix* is set to *_bd,* and *unicast_routing* is set to *false.*

`defaults.nac.yaml`

```yaml
defaults:
  apic:
    tenants:
      bridge_domains:
        name_suffix: "_bd"
        unicast_routing: false
```

`lab_tenant.nac.yaml`

```yaml
apic:
  tenants:
    - name: LAB_TENANT
      vrfs:
        - name: LAB_VRF
      bridge_domains:
        - name: vlan_1001
          vrf: LAB_VRF
        - name: vlan_1002
          vrf: LAB_VRF
        - name: vlan_1003
          vrf: LAB_VRF
```

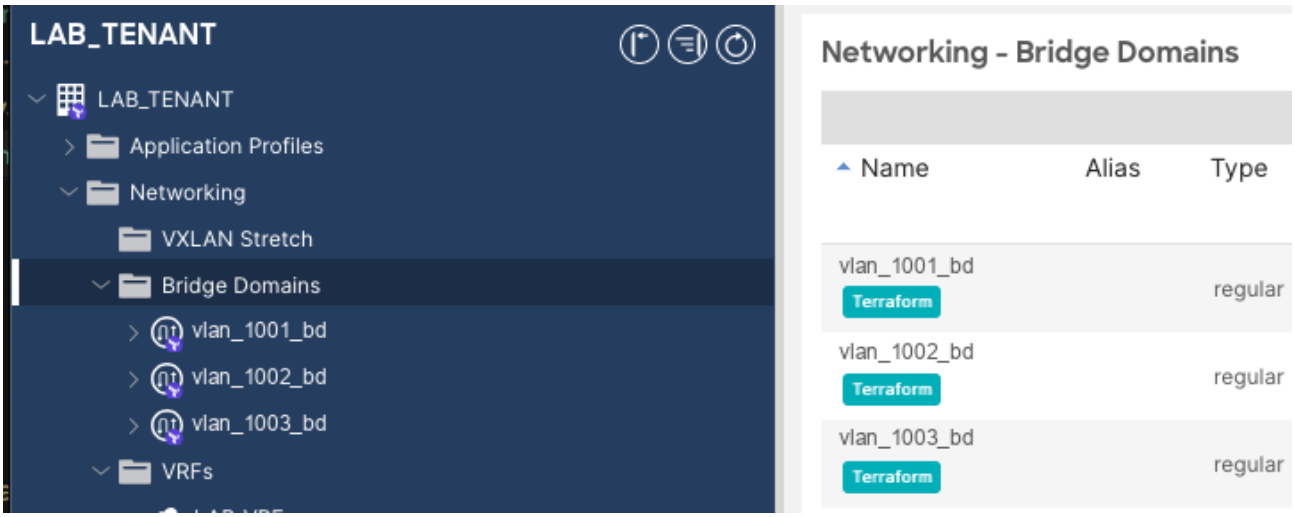
Before running the plan, set `manage_tenants = true` in `main.tf`. In the examples so far it was set to false, so Terraform would ignore the tenant configuration and the plan would show no tenant changes.

Running terraform plan against this configuration confirms that both overrides are applied automatically to every bridge domain in the data model, without needing to specify them individually on each object. The plan

output shows the rendered fvBD objects, with the suffix appended consistently to each bridge domain name, exactly as defined once in the defaults file.

```
  # module.aci.module.aci_bridge_domain["LAB_TENANT/vlan_1003"].aci_rest_managed.fvBD will be created
  + resource "aci_rest_managed" "fvBD" {
      + annotation  = "orchestrator:terraform"
      + class_name  = "fvBD"
      + content     = {
          + "arpFlood"             = "yes"
          + "descr"                = null
          + "epClear"              = "no"
          + "epMoveDetectMode"     = null
          + "hostBasedRouting"     = "no"
          + "ipLearning"           = "yes"
          + "limitIpLearnToSubnets" = "yes"
          + "mac"                  = "00:22:BD:F8:19:FF"
          + "mcastARPDrop"         = null
          + "mcastAllow"           = "no"
          + "multiDstPktAct"       = "bd-flood"
          + "name"                 = "vlan_1003_bd"
          + "nameAlias"            = null
          + "type"                 = "regular"
          + "unicastRoute"         = "no"
          + "unkMacUcastAct"       = "flood"
          + "unkMcastAct"          = "flood"
          + "v6unkMcastAct"        = "flood"
          + "vmac"                 = "not-applicable"
        }
      + dn          = "uni/tn-LAB_TENANT/BD-vlan_1003_bd"
      + escape_html = true
      + id          = (known after apply)
    }
```

Configuration pushed to the APIC further verifies that the suffix is appended to all Bridge Domain names as desired.



This illustrates the core value of default values in Network as Code: common conventions, such as naming suffixes, are declared a single time and propagate automatically to every matching object and its references, removing a recurring source of manual, error-prone repetition.

Terraform State

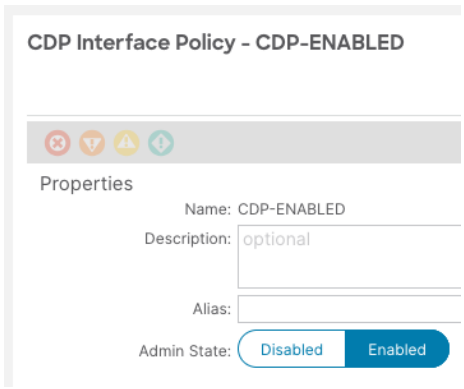
Following the initial deployment, a Terraform statefile (terraform.tfstate) is created.

A Terraform state file (terraform.tfstate) is a JSON file Terraform writes after every apply action. It is Terraform's record of what it believes exists, i.e. every resource it created, along with the attributes it set on each one, mapped to the real object IDs in the target system (in this case, the APIC).

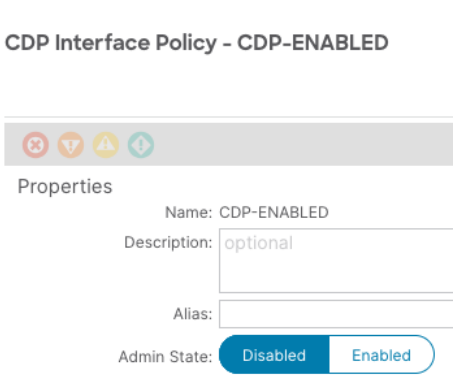
The YAML data model remains the source of truth for the intended configuration, while the state file is Terraform's record of what it manages. To show how Terraform uses both, a manual change was introduced directly in the APIC GUI to an object originally created by Terraform. The admin_state of the CDP-ENABLED interface policy was toggled from enabled to disabled through the APIC UI, outside of the Network as Code workflow.

Terraform change

```
      cdp_policies:
        - name: CDP-ENABLED
          admin_state: true
```



Manual override on the GUI



The Terraform state file was not automatically updated to reflect this change, since the manual change bypassed Terraform entirely. Running terraform plan afterward detects a deviation.

```
nac-aci-terraform % terraform plan

….

Terraform used the selected providers to generate the following execution plan. Resource
actions are indicated with the following symbols:

  ~ update in-place

Terraform will perform the following actions:

  # module.aci.module.aci_cdp_policy["CDP-ENABLED"].aci_rest_managed.cdpIfPol will be
updated in-place

  ~ resource "aci_rest_managed" "cdpIfPol" {
      ~ content     = {
          ~ "adminSt" = "disabled" -> "enabled"
            # (1 unchanged element hidden)
        }
        id          = "uni/infra/cdpIfP-CDP-ENABLED"
        # (4 unchanged attributes hidden)
    }
Plan: 0 to add, 1 to change, 0 to destroy.
```

Terraform refreshed the object's current state from the APIC, compared it with the state file, and then compared the YAML configuration against it. The APIC reports admin_state as disabled while the YAML declares it as enabled, so Terraform proposed reverting it to enabled so that the infrastructure matches the YAML. This three-way comparison between the YAML (desired state), the state file (what Terraform last applied) and the APIC

(actual state) shows that the YAML, not the live infrastructure, is the source of truth. The difference appears because the YAML declares admin_state: true, not because of anything stored in the state file.

Note: In this lab, the state file lives on the engineer's local machine, inside the nac-aci-terraform project folder. That is acceptable for learning the workflow, but it does not scale to a team, and it does not belong in a production process: a single laptop becomes a single point of failure, holding the only copy of the state, which no one else can see or safely edit. Once this workflow moves into Git and a CI/CD pipeline, covered in the next article, the state file no longer lives on any one engineer's machine. It is stored in a remote backend, in this case GitLab's managed Terraform state backend, so that the pipeline itself, not an individual laptop, always reads and writes one shared state for the fabric.

The next section of this lab will go into the Testing and Validation framework that is a key element of Network as Code.

Testing and Validation – Network as Code

Network as Code introduces a complete engineering lifecycle, with validation before deployment and testing after deployment. This is one of the major advantages of the Cisco Network as Code framework because rather than simply generating Terraform code and applying changes to Cisco ACI, the framework introduces validation throughout the entire infrastructure lifecycle. NaC offers tools for automated pre-deployment validation and post-deployment validation.

Pre-Change Validation

Pre-change/pre-deployment validation is the process within the engineering lifecycle whereby a proposed change is validated before any configuration has touched the infrastructure. Network as code offers a CLI tool called nac-validate for pre-change validation. The nac-validate tool is executed before configuration deployment to validate that the configuration files do not contain any syntax, semantic or logical errors. The intention is to fail fast: errors are caught at this pre-change stage, rather than waiting for the configuration to fail during the maintenance window.

The nac-validate tool can be found here: https://github.com/netascode/nac-validate

Before infrastructure changes are deployed**, nac-validate** analyzes the desired configuration to ensure it satisfies a series of increasingly sophisticated validation stages. Instead of waiting for Cisco ACI to reject invalid objects or generate faults after deployment, many issues can be detected early within the lifecycle.

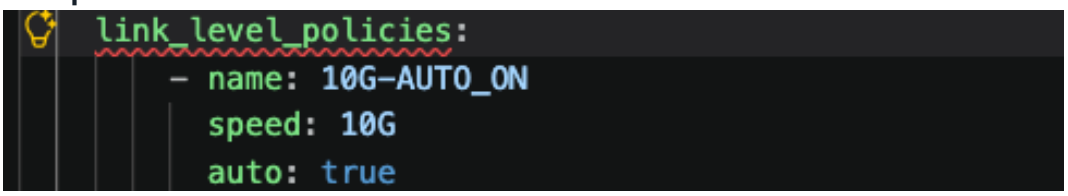
The pre-change validation process typically focuses on the following core areas:

- **Format Validation** – verifies the YAML documents are correctly formatted.
- **Syntax Validation** – ensures the YAML conforms to the expected schema and data types.
- **Semantic Validation** – validates logical consistency and referential integrity between objects.
    - *Naming standards*
    - *IP address format*
    - *Overlapping subnets*
    - *Complex validations across multiple objects*
- **Compliance Validation** – verifies an organization-specific policies and governance requirements.

Each stage examines the configuration from a different perspective and progressively increases confidence that the proposed change is safe to deploy. Gone are the days when “let’s see what happens” counted as pre-change validation.

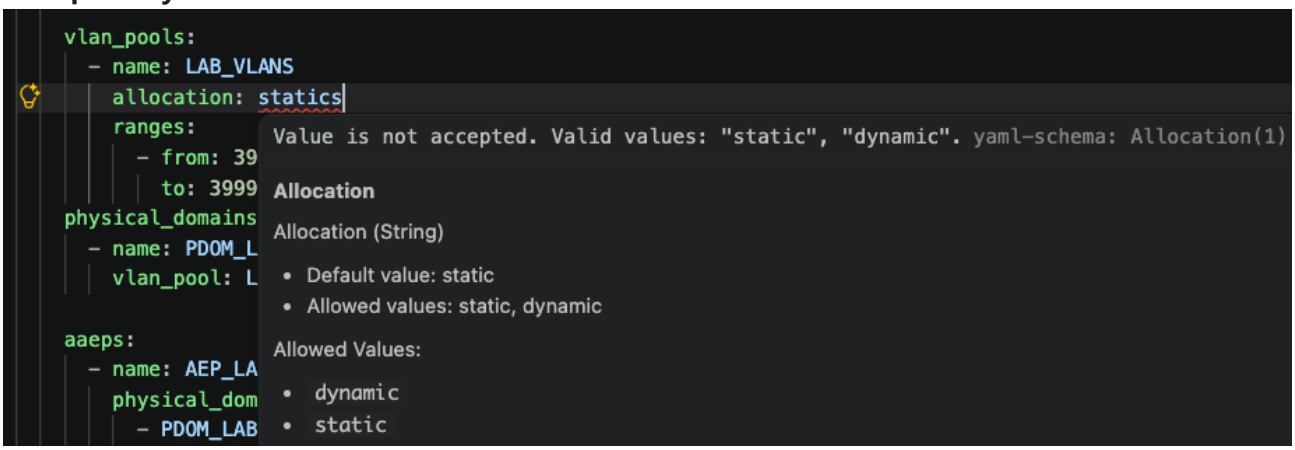
Let's walk through some examples to see pre-change validation in action during the YAML (intent) development stage. This is before the **"nac-validate"** CLI tool is officially executed to provide validation across all YAMLs. The checks in these examples are provided by the Visual Studio Code YAML plugin, which validates files with the .nac.yaml extension against the Network as Code data model as you type. They are unrelated to, and independent of, the nac-validate tool.

Example 1: YAML Format validation



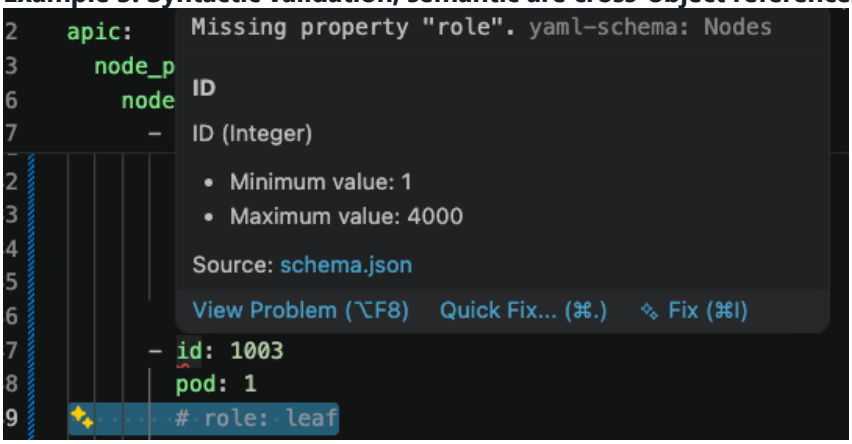
Any indentation violations are detected allowing the engineer to refine the YAML format.

Example 2: Syntactic schema validation



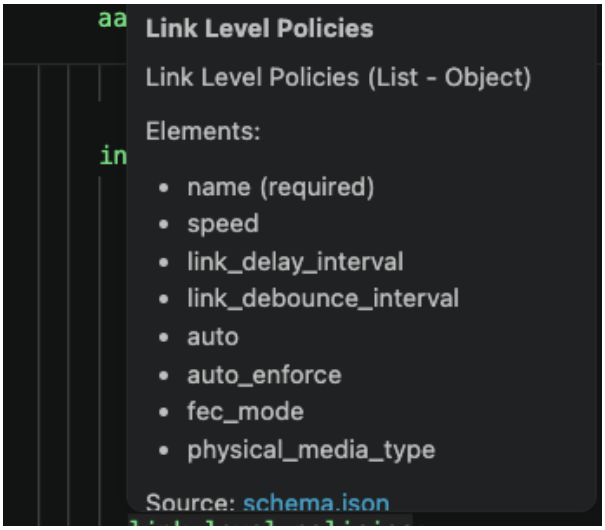
In this example, an incorrect value was put in a YAML file and this violation is already highlighted because specific keys have specific values that they accept & Network as code ensures that these validations are adhered to.

Example 3: Syntactic validation, semantic are cross-object references

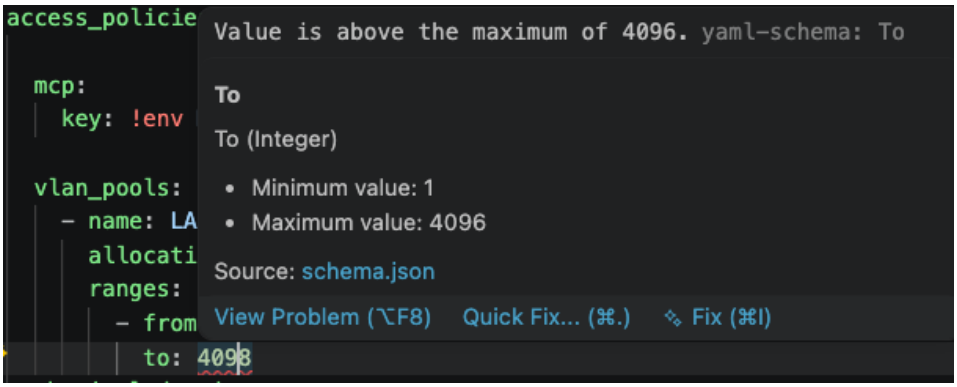


In this example, the key:value (role: leaf) which is a mandatory attribute was removed. This was already flagged to the user that there is a missing property. This is quite key because mistakes are caught way earlier within the lifecycle.

Example 4: Objects Validations against the schema



Example 5: Invalid VLAN ID in the data model



In this example a VLAN ID (4098). was defined which goes beyond the boundaries of the maximum VLAN ID (4096). The pre-validation tool flags this immediately.

Since the files are defined with the ".nac.yaml" extension, this offers a help context during the YAML development. Just by hovering over a property, in this case "link_level_policies: " the user can see the attributes that are required/allowed within that specific section of the configuration. This allows the user to quickly build up a valid intent that does not violate the data model.

Now let's run the "nac-validate" CLI tool to see it in action. Unlike the Visual Studio Code plugin, which runs on a laptop and helps while the YAML files are being created, nac-validate performs a comprehensive check of all the files, covering both syntax and semantics (semantic validation is not done by the plugin), and can be executed on any machine, including the one running the pipeline.

Note: The `data/` folder, where the YAML files are defined, is freely available (for example, in the nac-aci-comprehensive-example repository). The `.schema.yaml` file and the `.rules/` directory used in this command are not publicly available. They are provided to customers with a Services as Code subscription and are placed in the project root. Cisco Services as Code is available through Cisco Professional Services as an annual subscription service. To get started, contact your account representative or authorized Cisco partner today (Services as Code At-a-Glance).

The following CLI command is executed

```
nac-validate data/ -s .schema.yaml -r .rules/ --format json
```

The above command:

1. Reads all YAML input files under data/ folder and builds one inventory model.

2. Checks structure/types with *.schema.yaml* (required fields, value types, ranges, regex patterns, valid nesting).
3. Runs custom Python rules in *.rules/* (business logic checks like references, uniqueness, cross-section consistency).
4. Collects all findings into categories like schema/syntax vs semantic rule violations.
5. Prints the result as JSON because of --format json.

Note:

- The .schema.yaml is a schema definition that describes the expected shape and types of the data model. It does not perform any checks itself: nac-validate uses it to verify that fields exist where required, that values are the right type/range/format, and that nesting matches the model.

- The .rules/*.py files do semantic and cross-object validation: logic that schema alone can't express cleanly, like uniqueness across lists, reference integrity between sections, and custom policy behavior.

- The `--format json` option is optional and a matter of personal preference. Without it, nac-validate prints its default plain CLI output, which works just as well.

To test the nac-validate towards a configuration rule violation, I have defined a data model (intent) which assigns a fabric_policy_group attribute to the nodes (leafs and spines) however without this policy group defined anywhere else.

```
---
apic:
  node_policies:
    nodes:
      - id: 1001
        pod: 1
        role: leaf
        serial_number: FDXXXXXXXX
        name: leaf1
        oob_address: 10.x.y.232/27
        oob_gateway: 10.x.y.225
        fabric_policy_group: ALL_LEAFS
        access_policy_group: ALL_LEAFS
```

The nac-validate is able to flag this rule violation, allowing an engineer to make the appropriate object references before pushing the configuration to the controller.

```
nac-aci-terraform % nac-validate data/ -s .schema.yaml -r .rules/ --format json
{
  "syntax_errors": [],
  "semantic_errors": [
    {
      "rule_id": "202",
      "description": "Verify Fabric Leaf Switch Policy Group references",
      "severity": "HIGH",
      "errors": [
        {
          "message": "apic.node_policies.nodes.fabric_policy_group - ALL_LEAFS"
        },
        {
          "message": "apic.node_policies.nodes.fabric_policy_group - ALL_LEAFS"
        },
        {
```

```
          "message": "apic.node_policies.nodes.fabric_policy_group - ALL_LEAFS"
        },
        {
          "message": "apic.node_policies.nodes.fabric_policy_group - ALL_LEAFS"
        }
      ]
    },
    {
      "rule_id": "203",
      "description": "Verify Fabric Spine Switch Policy Group references",
      "severity": "HIGH",
      "errors": [
        {
          "message": "apic.node_policies.nodes.fabric_policy_group - ALL_SPINES"
        },
        {
          "message": "apic.node_policies.nodes.fabric_policy_group - ALL_SPINES"
        }
      ]
    }
  ]
}
```


The fix for this requires the definition of the missing fabric_policy group to be defined along with its relevant policies.

main.tf

```
module "aci" {
  source  = "netascode/nac-aci/aci"
  version = "2.0.0"

  yaml_directories = ["data"]
  manage_access_policies    = true
  manage_fabric_policies    = true
  manage_pod_policies       = true
  manage_node_policies      = true
  manage_interface_policies = true
  manage_tenants            = false

  write_default_values_file = "defaults.yaml"
}
```

fabric_policies.nac.yaml

```
apic:
  fabric_policies:

    switch_policies:
      psu_policies:
        - name: COMBINED
          admin_state: combined

    leaf_switch_policy_groups:
      - name: ALL_LEAFS
        psu_policy: COMBINED


    spine_switch_policy_groups:
      - name: ALL_SPINES
        psu_policy: COMBINED
```

Now that the fabric spine/switch policy groups have been defined, in this example with a power supply policy (optional) associated with the groups, the nac-validate will be executed again to validate if there are any missing references in the data model.

```
nac-aci-terraform % nac-validate data/ -s .schema.yaml -r .rules/

────────────────────────────────────────────────────────────────────────────────

Validation Summary

────────────────────────────────────────────────────────────────────────────────

  ✓ Syntax validation: PASSED (7 files)
  ✓ Semantic validation: PASSED
```

Network as code allows for custom rules/policy enforcement whereby an organization can define its own objects naming conventions, IP addressing schemes, security and regulatory compliance etc.

In the example below, a simple custom naming convention rule was created which only accepts the name “BFD-IPV4-POLICY” when defining a BFD IPv4 policy. To test that the rule is indeed enforced, the BFD policy name before has been defined as “BFD-POLICY” which does not conform to the required convention.

access_policies.nac.yaml

```yaml
apic:
  access_policies:
    switch_policies:
      bfd_ipv4_policies:
        - name: BFD-POLICY
          description: BFD IPv4 Policy
          detection_multiplier: 5
          min_transmit_interval: 100
```

When the nac-validate is executed, the rule enforcement is triggered:

```
nac-aci-terraform % nac-validate data/ -s .schema.yaml -r .rules/ --format json
{
  "syntax_errors": [],
  "semantic_errors": [
    {
      "rule_id": "500",
      "description": "Verify BFD IPv4 Policy naming convention",
      "severity": "HIGH",
      "errors": [
        {
          "message": "apic.access_policies.switch_policies.bfd_ipv4_policies.name - BFD-POLICY
does not follow the naming convention. Name must be 'BFD-IPV4-POLICY'"
        }
      ]
    }
  ]
}
```

After the fix, has been put in place in the data model, all pre-validation checks pass.

access_policies.nac.yaml

```yaml
apic:
  access_policies:
    switch_policies:
      bfd_ipv4_policies:
        - name: BFD-IPV4-POLICY
          description: BFD IPv4 Policy
          detection_multiplier: 5
```

nac-validate checks

```
nac-aci-terraform %
nac-validate data/ -s .schema.yaml -r .rules/
──────────────────────────────────────────────────

Validation Summary
──────────────────────────────────────────────────

  ✓ Syntax validation: PASSED (7 files)
  ✓ Semantic validation: PASSED
```

This pre-validation summary result provides the exact confidence required before pushing configuration to the infrastructure.

Post-Change Validation

This section introduces the post-change validation framework and tool offer by Network as Code. There are certain aspects that can only be verified/tested after configuration has been pushed to the environment e.g. the actual operational state. Network as Code has a specific tool for post-change validation which is nac-test.

The nac-test tool can be found here: https://github.com/netascode/nac-test

Note: nac-test itself is publicly available. The test templates it renders and executes (referenced later with the `-t tests/templates` option) are not publicly available and require a Services as Code subscription (Services as Code At-a-Glance). The examples in this section show what is possible with the subscription.

A successful configuration deployment is certainly a win. However, the reality is that successfully pushing configuration does not necessarily mean the network has reached its intended operational state. The APIC may have accepted the configuration, but that alone does not prove that the network is actually working as intended.

This is where post-change validation becomes critical. Network as Code does not treat validation as an afterthought. The nac-test tool provides a way to automate the verification of the network after configuration has been deployed, allowing the intended configuration as part of the same automation lifecycle.

nac-test can render and execute automated tests using both Robot Framework and pyATS, with test suites driven by the same YAML data model used to describe the network intent. This enables validation to move beyond simply asking "Did the configuration deploy?" toward the far more important question:

"Did the network actually become what we intended it to become?"

The specific test cases that are executed are not generic as they are dynamically rendered based on the defined data model. The set of post-change test cases can be classified under the following categories.

Configuration Tests
Verify that the desired configuration has been successfully rendered within Cisco ACI.

Health Tests
Verify that the infrastructure remains healthy following the deployment and that no new faults or degradation have been introduced.

```
nac-aci-terraform % nac-test -d ./data -d ./defaults.yaml -t tests/templates -f tests/filters -o tests/results
📄 Merging data model files...
✅ Data model merging completed (< 1s)

🤖 Running Robot Framework tests...

📝 Rendering Robot Framework templates...
🤖 Executing Robot Framework tests...
… output truncated

2026-08-08 12:16:38.395904 [PID:2606] [6] [ID:48] PASSED Robot Results.Config.Node Policies.Vpc Group in 1.6 seconds
2026-08-08 12:16:38.961951 [PID:2604] [4] [ID:46] PASSED Robot Results.Config.Node Policies.Node Registration in 2.2
seconds
2026-08-08 12:16:38.992718 [PID:2605] [5] [ID:47] PASSED Robot Results.Config.Node Policies.Oob Node Address in 2.2
seconds
140 tests, 140 passed, 0 failed, 0 skipped.
```

```
==============================================
Output:  nac-aci-terraform/tests/results/robot_results/output.xml
XUnit:   nac-aci-terraform/tests/results/robot_results/xunit.xml
Log:     nac-aci-terraform/tests/results/robot_results/log.html
Report:  nac-aci-terraform/tests/results/robot_results/report.html
Finalizing Pabot execution...
Stopping PabotLib process
Robot Framework remote server at 127.0.0.1:60373 stopped.
PabotLib process stopped
Total testing: 1 minute 32.80 seconds
Elapsed time:  8.77 seconds
Logs flushed successfully.

======================================================================
Combined Test Execution Summary
----------------------------------------------------------------------
140 tests, 140 passed, 0 failed, 0 skipped.
----------------------------------------------------------------------
Dashboard:  nac-aci-terraform/tests/results/combined_summary.html
Robot:      nac-aci-terraform/tests/results/robot_results/log.html
xUnit:      nac-aci-terraform/tests/results/xunit.xml
```

A note on *defaults.yaml (referenced in the nac-test command)*

The nac-test command above references two data sources: ./data, which holds the intent-based configuration you author, and ./defaults.yaml, which supplies the module's effective default values for anything you did not explicitly override.

The *defaults.yaml* is generated automatically by Terraform as a side effect of terraform apply, via the module's write_default_values_file setting (configured in this project's main.tf). It captures the built-in defaults defined by the nac-aci module, merged with any overrides that were made in data/defaults.nac.yaml. Test templates validate configuration against the values that were applied; including defaulted ones, so nac-test needs both the explicit intent and the resolved defaults to build accurate test assertions.

Because this file is generated, it is only as current as your last apply. It should be regenerated whenever the module version changes, so newly introduced default fields stay in sync.

The nac-test produces artifacts that are easy to consume. A log.html is one of the most important artifacts that one can go through to analyze the post-validation result.



Robot Results Log

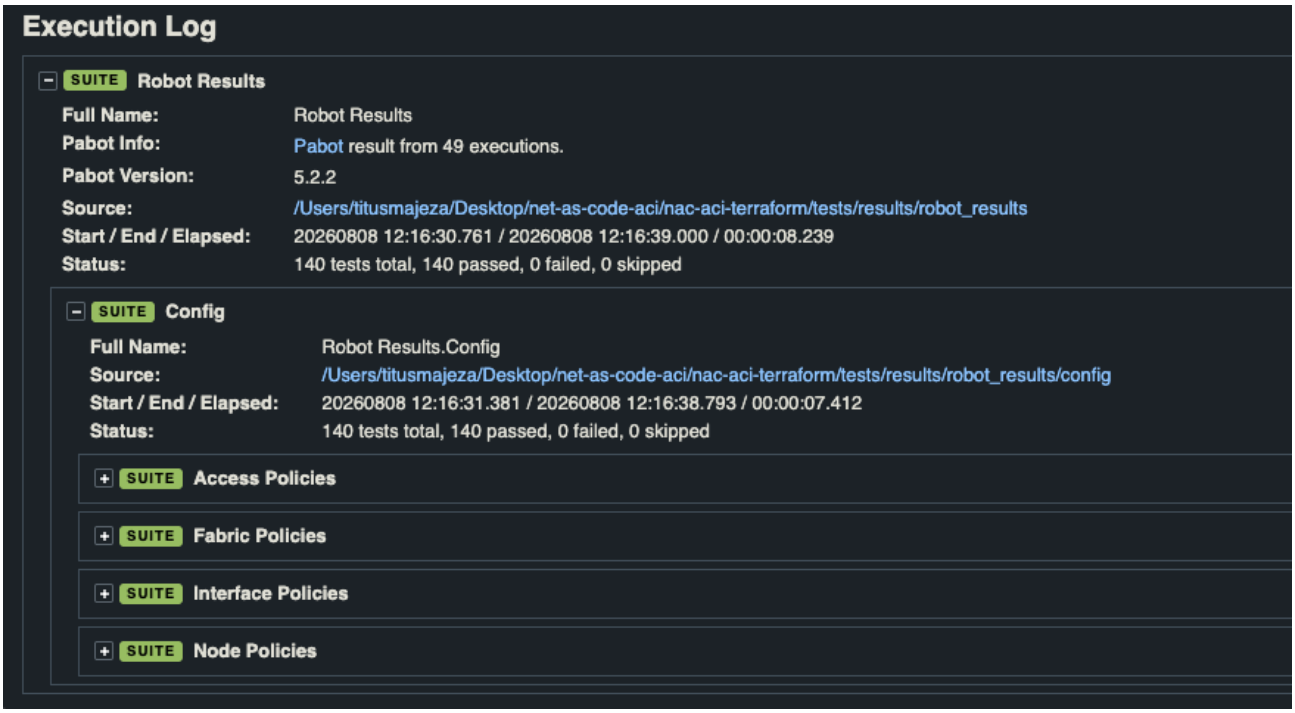
Generated
20260808 12:16:39 UTC+02:00
6 minutes 38 seconds ago

Statistics

Total Statistics	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
All Tests	140	140	0	0	00:00:42	

Statistics by Tag	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
access_policies	55	55	0	0	00:00:16	
apic	140	140	0	0	00:00:42	
config	140	140	0	0	00:00:42	
day0	35	35	0	0	00:00:15	
day1	102	102	0	0	00:00:25	
day2	3	3	0	0	00:00:02	
fabric_policies	68	68	0	0	00:00:20	
interface_policies	2	2	0	0	00:00:01	
node_policies	15	15	0	0	00:00:04	

Statistics by Suite	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
Robot Results	140	140	0	0	00:00:08	
Robot Results . Config	140	140	0	0	00:00:07	
Robot Results . Config . Access Policies	55	55	0	0	00:00:04	
Robot Results . Config . Access Policies . Aaep	2	2	0	0	00:00:02	
Robot Results . Config . Access Policies . Ap Leaf Interface Policy Group	1	1	0	0	00:00:02	
Robot Results . Config . Access Policies . Ap Leaf Interface Profile	4	4	0	0	00:00:04	
Robot Results . Config . Access Policies . Ap Leaf Switch Policy Group	1	1	0	0	00:00:02	
Robot Results . Config . Access Policies . Ap Leaf Switch Profile	20	20	0	0	00:00:04	
Robot Results . Config . Access Policies . Ap Spine Interface Profile	2	2	0	0	00:00:03	
Robot Results . Config . Access Policies . Ap Spine Switch Policy Group	1	1	0	0	00:00:03	
Robot Results . Config . Access Policies . Ap Spine Switch Profile	8	8	0	0	00:00:04	
Robot Results . Config . Access Policies . Bfd Policy	1	1	0	0	00:00:02	
Robot Results . Config . Access Policies . Cdp Policy	1	1	0	0	00:00:03	
Robot Results . Config . Access Policies . Link Level Policy	1	1	0	0	00:00:03	
Robot Results . Config . Access Policies . Lldp Policy	1	1	0	0	00:00:03	
Robot Results . Config . Access Policies . Mcp	1	1	0	0	00:00:02	
Robot Results . Config . Access Policies . Physical Domain	2	2	0	0	00:00:03	
Robot Results . Config . Access Policies . Qos	7	7	0	0	00:00:01	
Robot Results . Config . Access Policies . Vlan Pool	2	2	0	0	00:00:01	
Robot Results . Config . Fabric Policies	68	68	0	0	00:00:03	



Execution Log

SUITE **Robot Results**

Full Name:	Robot Results
Pabot Info:	Pabot result from 49 executions.
Pabot Version:	5.2.2
Source:	/Users/titusmajeza/Desktop/net-as-code-aci/nac-aci-terraform/tests/results/robot_results
Start / End / Elapsed:	20260808 12:16:30.761 / 20260808 12:16:39.000 / 00:00:08.239
Status:	140 tests total, 140 passed, 0 failed, 0 skipped

SUITE **Config**

Full Name:	Robot Results.Config
Source:	/Users/titusmajeza/Desktop/net-as-code-aci/nac-aci-terraform/tests/results/robot_results/config
Start / End / Elapsed:	20260808 12:16:31.381 / 20260808 12:16:38.793 / 00:00:07.412
Status:	140 tests total, 140 passed, 0 failed, 0 skipped

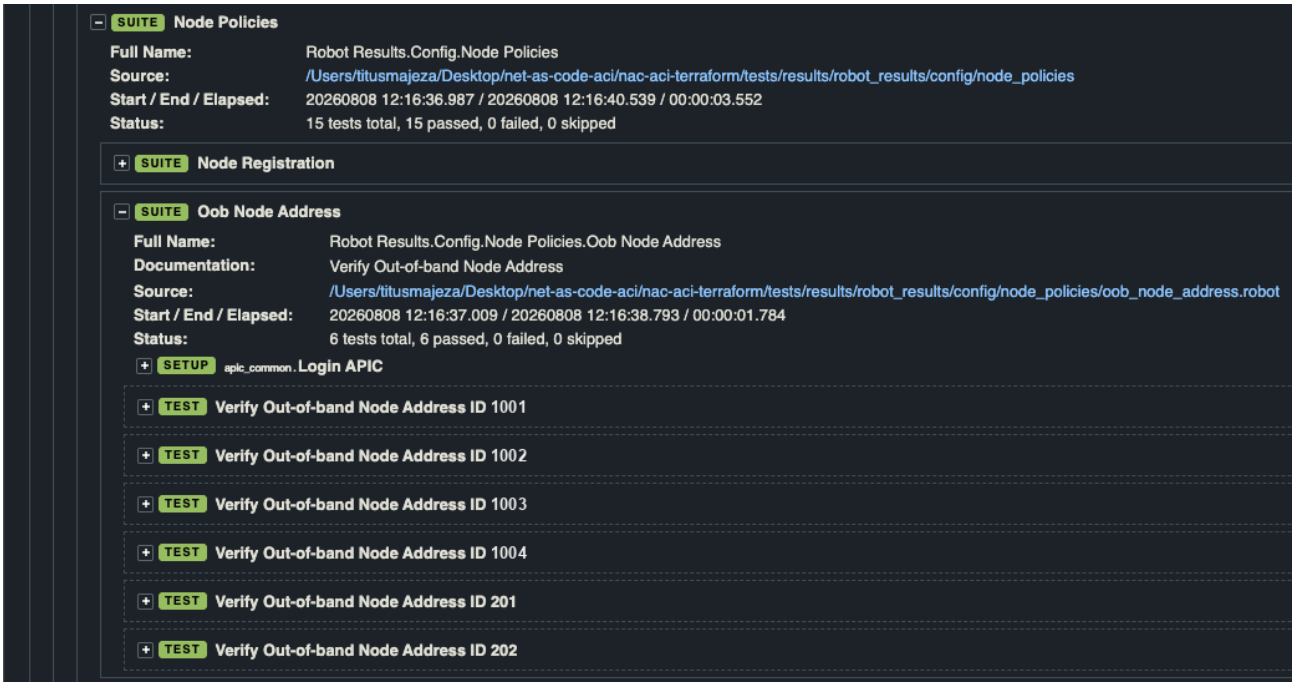
SUITE **Access Policies**

SUITE **Fabric Policies**

SUITE **Interface Policies**

SUITE **Node Policies**

The HTML offers a high-level summary of test cases that were executed and an interactive capability for the user to dig in a specific Suite for low level details of each test case that was executed.



Catching Configuration Drifts

To validate that nac-test catches configuration drift, the admin_state field of the CDP-ENABLED interface policy in *data/access_policies.nac.yaml* was changed from true to false, without running terraform apply. This introduced a deliberate mismatch: the data model now declared CDP as disabled, while the APIC still had it enabled from the last actual deployment. Editing the YAML is a simple, lightweight way to simulate drift, and a softer demonstration than a real out-of-band change on the APIC, which is described below.

```
nac-aci-terraform % nac-test -d ./data -d ./defaults.yaml -t tests/templates -f tests/filters -o
tests/results
📄 Merging data model files...
✅ Data model merging completed (< 1s)

🤖 Running Robot Framework tests...

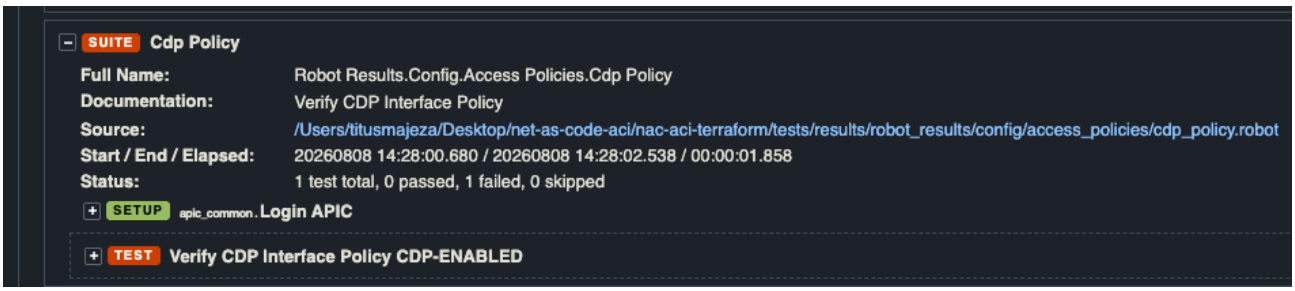
📝 Rendering Robot Framework templates...
🤖 Executing Robot Framework tests...

… output truncated

2026-08-08 13:21:20.203232 [PID:75703] [4] [ID:9] Execution failed in Robot Results.Config.Access
Policies.Cdp Policy with 1 failing test(s)
==============================================================================
Robot Results
==============================================================================
Robot Results.Config
==============================================================================
Robot Results.Config.Access Policies
==============================================================================
Robot Results.Config.Access Policies.Cdp Policy :: Verify CDP Interface Policy
==============================================================================
Verify CDP Interface Policy CDP-ENABLED                               | FAIL |
enabled != disabled
------------------------------------------------------------------------------
Robot Results.Config.Access Policies.Cdp Policy :: Verify CDP Inte... | FAIL |
1 test, 0 passed, 1 failed
==============================================================================
Robot Results.Config.Access Policies                                  | FAIL |
1 test, 0 passed, 1 failed
==============================================================================
Robot Results.Config                                                  | FAIL |
1 test, 0 passed, 1 failed
==============================================================================
```

```
Robot Results                                                         | FAIL |
1 test, 0 passed, 1 failed
==============================================================================


==========================================================================
Combined Test Execution Summary
--------------------------------------------------------------------------
140 tests, 139 passed, 1 failed, 0 skipped.
--------------------------------------------------------------------------
```



Running the nac-test against this state produced a failure on the Configuration Test for that policy, since the test validated the fabric against the current data model rather than what was last applied. The result confirmed that post-change validation correctly surfaces drift between declared intent and deployed state.

In practice, the more common source of drift is an out-of-band change: i.e. someone logs in to the APIC and changes a setting manually. This is the same situation shown in the Terraform State section, where the `admin_state` of the CDP-ENABLED policy was toggled in the APIC GUI. Because the data model still declares the intended value, running nac-test would flag the mismatch on the corresponding configuration test and identify exactly which object differs, even though nothing was changed in the YAML files. Catching these manual changes is what makes post-change validation valuable beyond a single deployment.

Version Control & CI Pipeline

Every YAML file in this data model is plain text, which means it can be tracked in Git. Every change becomes visible, reviewable, and reversible, turning the repository into a logbook of why, when, and by whom the change was made. The next article picks this up in full, version controlling this data model in Git, and wiring that repository into a CI/CD pipeline so every change is automatically validated, tested, and deployed without a manual terraform apply run from a laptop.

Start Your Network as Code Journey Today

To quickly get started with Network as code for ACI deployment and validation, you can clone either of the following repositories.

- **Simple Example:** https://github.com/netascode/nac-aci-simple-example
- **Comprehensive Example:** https://github.com/netascode/nac-aci-comprehensive-example

Both provide a preconfigured directory structure with example configurations, giving you a ready-to-use starting point that can be modified to suit your environment.

References

1. **Network as Code official website**
   https://netascode.cisco.com

2. **Cisco Live Sessions:**
   - Deploy Network Changes with Confidence: Test and Validation with Network as Code - BRKOPS-1603
   - Network as Code - Kick-Start Your Automation with ACI - BRKDCN-2673
   - A Beginner's Guide to Network as Code for ACI - DEVNET-1202

3. **Cisco official Whitepapers**
   - https://www.cisco.com/c/en/us/solutions/collateral/executive-perspectives/technology-perspectives/automate-network-infrastructure-code-wp.html